Hardware User Manual

AnyDIO Series

SIO-DI32

AJINEXTEK AXT

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Revision History

Manual	PCB	Comments
Rev. 1.0 issue 1.0	Rev. 1.0	2001. 12.18

1. Outline

1.1.Introduction

AnyDIO module board is loaded on the AnyBus Series board. It has an input module, output module and input-output module which can control digital signal's input-output, and it is composed of various product group providing up to 32 points of IO contacts per each module. Among the models of AnyDIO module boards, the SIO-DI32 module board provides 32 points of digital input contacts.

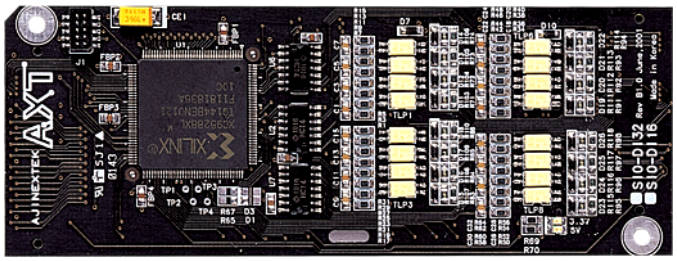
1.2.Application

AnyDIO module board can interface the external digital I/O signals, and can be used on the factory automation, machine tool control, laboratory work, educational purpose, etc.

- ▶ Digital Monitoring
- ▶ Signal Conversion
- ▶ Optical Insulation between Computer and the Peripheral
- ▶ Automatic Test Equipment
- ▶ Equipment Monitoring
- ▶ Equipment Interface
- ▶ Process Control
- ▶ Industrial Measurement
- ▶ Industrial Digital Input Control

2. Features

2.1.Module Specifications and Scheme



Picture 1. Real Picture of SIO-DI32 Module Board

Table 1. Features of SIO-DI32 Module Board

Items	SIO-DI32
Input	32 points
	32 points input can be accessed in bit and byte units.
	If a random bit inside of input bits was activated, a special interrupt is generated and the location where interrupt generated can be identified.
	Interrupt removal can be done by a library function.
Interrupt	Every Input
Range of Input Voltage	12~24VDC
Input Resistance	4.7kΩ
I/O Circuit	Sink (Negative logic)
	Every input port is electrically isolated with the external signal, so the external signals of 12V, 24V can be recognized or driven. Isolation makes use of a photo coupler.

Table 2. Specifications of SIO-DI32 Module Board

# of Input Channel	32 points
Input Range	External power license mode (12~24VDC)
Input Mode	Voltage
	Adopt noise prevention LPF per each input channel
Range of Input Voltage	Low (min : 0v ~ max : 12.0V)
	High (min : 12.5V ! max : 24V)
Input Load Current	Low (2mA – ON)
	High (2.1mA – OFF)
Active Indicator	2 LEDs for the license confirmation (5VDC, 3.3VDC)
Others	2 LEDs for the license confirmation
	Characteristics of isolation: 2500VRMS

3. Installation

3.1.Hardware Installation

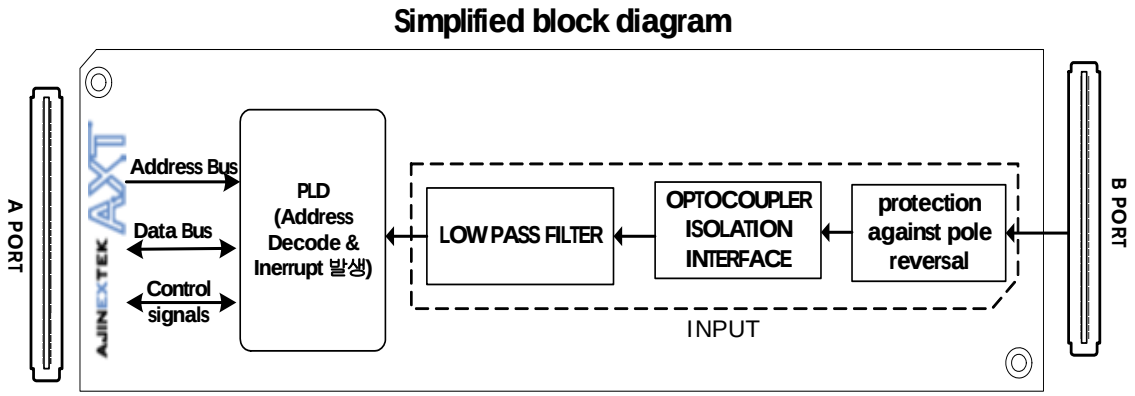
The loading direction of an AnyPack module (SIO-DI32) in Add-On type varies according to the basic mode baseboard a user selected. Load a module properly with reference to 'Hardware Description' of the baseboard user manual and then install according to the order of 'Hardware Installation'.

3.2.Software Installation

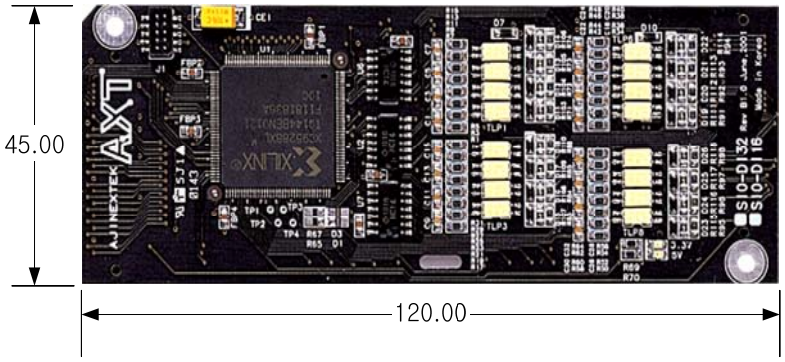
Install EzConfig referring to the EzConfig 1.0 user manual. If an installation target folder is not assigned yet, it is presumed that installation of EzDioAgent program is normally completed when the EzDioAgent.exe executive file exists in the C:\Program Files\EzConfig. If the installation target folder is assigned separately, check whether the EzDioAgent.exe executive file exists in the assigned folder or not.

4. Configuration

4.1.Hardware Configuration and Description



Picture 2. Block Diagram of SIO-DI32 Module Board



Picture 3. Size of SIO-DI32 Board

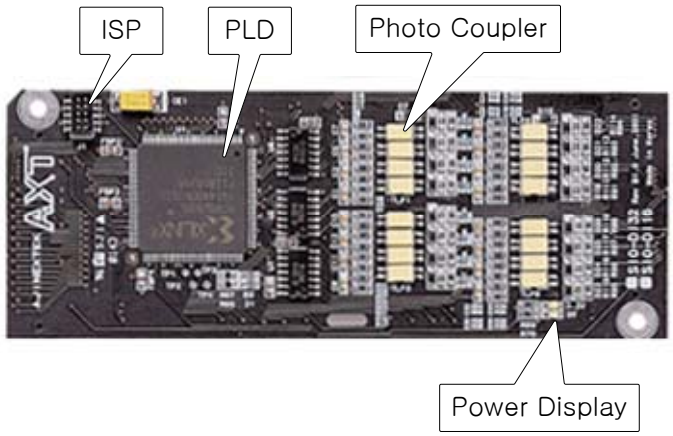
Table 3. Environment in Use of SIO-DI32 Module Board

Board Size	120 x 45 mm
Working Voltage	3.3V, 5V, 24V (12V

Depending on how you combine AnyDIO modules on the AnyBus board, the number of control IO contacts can be extended or modified by a simple operation in the control system using various IO devices.

If BPFR(PCI Full Size) is selected as a basic mode baseboard, it can compose up to 128 points with 4 AnyDIO

modules having 32 contacts. And it can be composed of 128 points input board, 64 points input/64 points output board and 128 points output board according to the selection of AnyDIO series. If SUB1 is used as a CPU slot on the BPFR board, it can get 96 DIO contacts by loading up to 32 points of AnyDIO on the 3 SUB slots. This configuration method can be changed into 96 points input (three SIO-DI32), 96 points output (three SIO-DO32P), 48 points input/48 points output (three SIO-DB32P), etc. depending on which AnyDIO module is used.

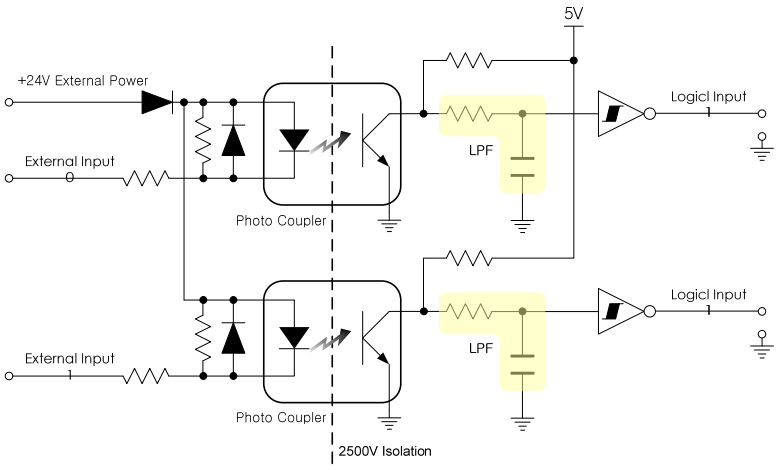


Picture 4. Configuration of SIO-DI32

Picture 4 shows a control division (PLD), SUB module connector, photo coupler, output drive, indication LED, etc. of the SIO-DI32.

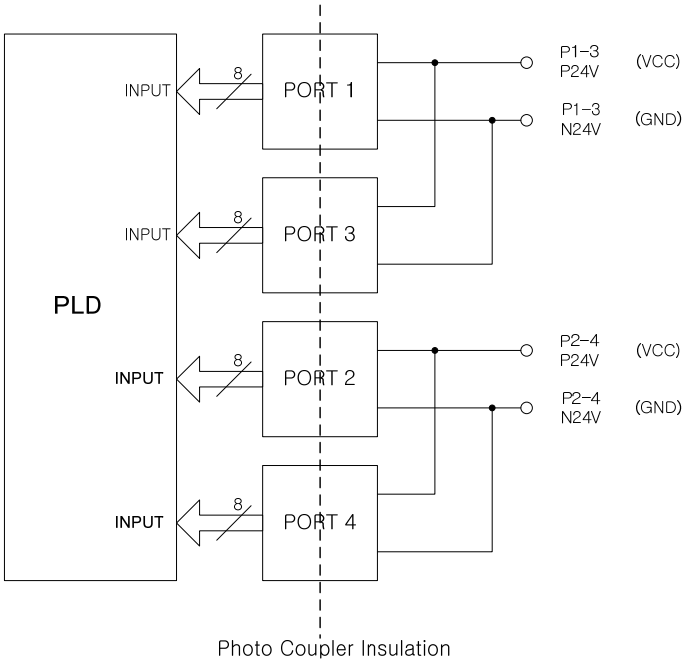
There are three wide functions in the control division. First, it has a function to read 32 points input. Second, it controls a data flow between the basic mode baseboard and the local module board. The last, it has an interrupt processing function generated from the local bus. Therefore users can control hardware easily. The first function implies that input control provides a buffer in order to read the external data of the basic mode baseboard so the external data can be received smoothly. Second function is that it separates registers by internal functions through the decoder so it makes data transmission between the basic mode baseboard and the local module easy. Lastly, the interrupt processing division can process the special bit only with an interrupt mask, and also process the change of state of the external input signal individually or simultaneously at the rising edge and the falling edge. It generates interrupt and transmits to the control division of the basic mode baseboard then it finally helps the user to apply.

64 pin PMC(PCI Mezzanine Card) SUB module connectors of the A port and the B port are summarized in short as follows. The A port supplies 5V and 3.3V power to module from the AnyBus baseboard. The signal line of the A port is composed of data 8 bit, address 9 bit, read and write signals, selection signal of module, one interrupt request signal (IRQ) and so on. The B port receives the external power supply and 32 input digital IO contacts are connected with the 68 CHAMP connector. For the input, photo couplers are generally used where grounds are hardly used in common or when sensors that send out the digital output are different from each other.



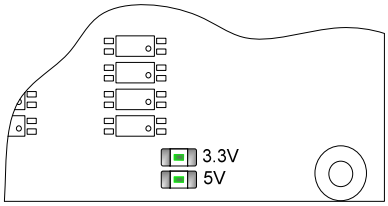
Picture 5. In Case of Using Input Signal's Photo Coupler

All the 32 points digital inputs are isolated with the photo coupler. Having a low pass filter (LPF) within each input circuit, it prevents malfunction or system damage from a noise or searching voltage and removes chattering generated when using mechanical contacts.



Picture 6. Power Processing inside of the Module

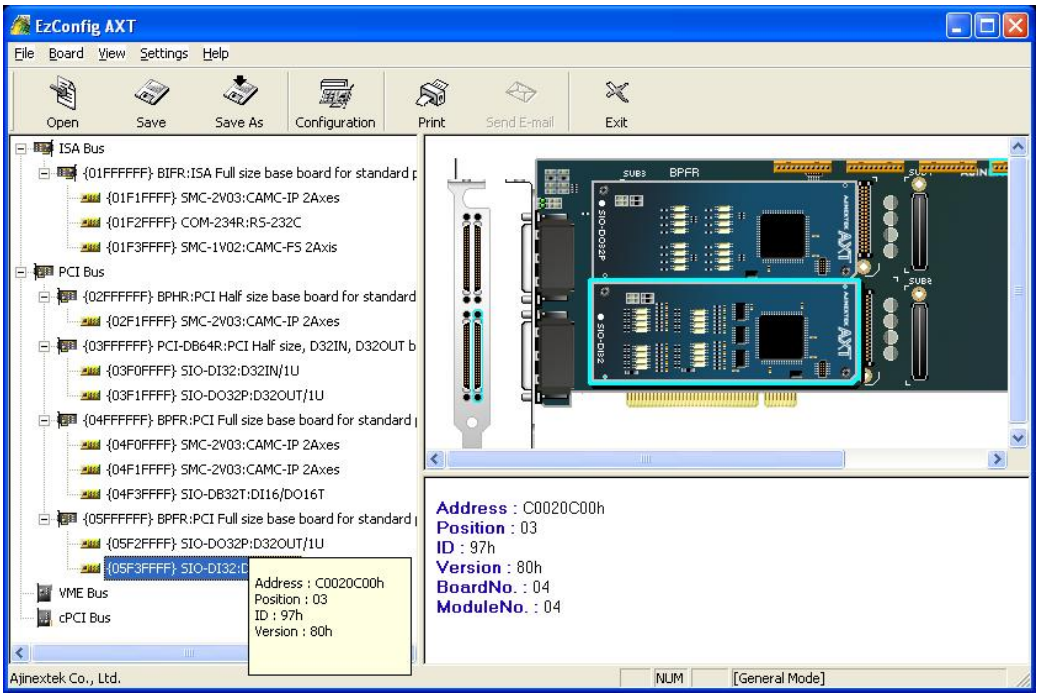
Picture 6 shows the power process inside of module. Each port is composed of 8 bits. Port1 and Port3 are connected with the same power supply line, and so are port2 and port4. Therefore it is easy to separate from equipment or sensor that makes a lot of noise by using each different power supplier.



Picture 7. Power Indication LED

Two LEDs on the SIO-DI32 module board are attached in order to indicate getting a supply of 3.3V and 5V power from the AnyBus baseboard. The LED is turned on if power is supplied from the AnyBus base and turned off if power is not supplied.

4.2.Module Address Map



Picture 8. SIO-DI32 Information View Screen Using EzConfig Software

Table 4. Range of Address Allocated to Each SUB

Region	BIFR, BV6R, BPFR, BC6R (FULL SIZE / 6U)	BIHR, BV3R, BPHR, BC3R (HALF SIZE / 3U)	Remarks
	0000h – 1FFFh	0000h – 1FFFh	
0000h – 03FFh	Address space allocated to SUB1	Address space allocated to SUB1	1Kbyte
0400h – 07FFh	Address space allocated to SUB2	Address space allocated to SUB2	1Kbyte
0800h – 0BFFh	Address space allocated to SUB3	-	1Kbyte
0C00h – 0FFFh	Address space allocated to SUB4	-	1Kbyte

If you select BIFR (ISA Full Size) board as a basic mode baseboard and start viewing information of SIO-DB32P module loaded on the SUB1 by using EzConfig software, you can get information that an address of module1 is the initial address of SIO-DI32.

Address = Base Address + 0000h(SUB 1 Offset Address)

If you press baseboard (B) tab on the information view dialog window, you can get information about the baseboard easily.

4.2.1.Address Map

Table 5. Address Map of SIO-DI32 Module Board

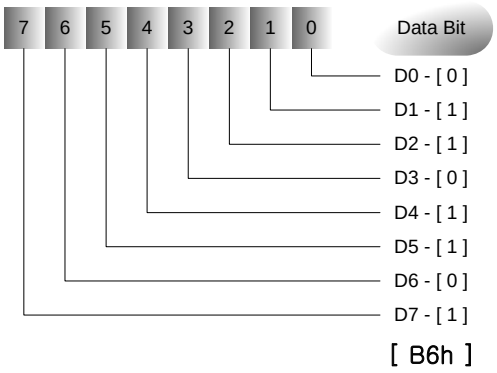
Offset	Function	Data	Length	Direction	Initial Value
00h	Preamble		8 bit	R	B6h
02h	ID		8 bit	R	97h (151)
04h	Version		8 bit	R	00h
06h	Software Reset	Bit0 : 1(ON) Software reset when Write	1 bit	R/W	00h
..0Fh	Reserved				
10h	Reserved				
12h	Reserved				
14h	Reserved				
16h	Reserved				
20h	Input port 1	Bit0..7 : 0(OFF)/1(ON)	8 bit	R	
22h	Input port 2	Bit0..7 : 0(OFF)/1(ON)	8 bit	R	
24h	Input port 3	Bit0..7 : 0(OFF)/1(ON)	8 bit	R	
26h	Input port 4	Bit0..7 : 0(OFF)/1(ON)	8 bit	R	
30h	Interrupt Flag port 1	Bit0..7 : 0(OFF)/1(ON) Clear when Write	8 bit	R	00h
32h	Interrupt Flag port 2	Bit0..7 : 0(OFF)/1(ON) Clear when Write	8 bit	R	00h
34h	Interrupt Flag port 3	Bit0..7 : 0(OFF)/1(ON) Clear when Write	8 bit	R	00h
36h	Interrupt Flag port 4	Bit0..7 : 0(OFF)/1(ON) Clear when Write	8 bit	R	00h
40h	Interrupt Rising edge port 1	Bit0..7 : 0=None / 1=Rising	8 bit	R/W	00h
42h	Interrupt Rising edge port 2	Bit0..7 : 0=None / 1=Rising	8 bit	R/W	00h
44h	Interrupt Rising edge port 3	Bit0..7 : 0=None / 1=Rising	8 bit	R/W	00h
46h	Interrupt Rising edge port 4	Bit0..7 : 0=None / 1=Rising	8 bit	R/W	00h
50h	Interrupt Falling edge port 1	Bit0..7 : 0=None / 1=Falling	8 bit	R/W	00h
52h	Interrupt Falling edge port 2	Bit0..7 : 0=None / 1=Falling	8 bit	R/W	00h
54h	Interrupt Falling edge port 3	Bit0..7 : 0=None / 1=Falling	8 bit	R/W	00h
56h	Interrupt Falling edge port 4	Bit0..7 : 0=None / 1=Falling	8 bit	R/W	00h
60h	Interrupt Enable control	Bit0 : 1(ON)=Enable 0(OFF)=Disable	1 bit	R/W	00h
62h	Interrupt status	Bit0 : 1(ON)=Occur, Automatic Clear when Clear the whole Flag Register	1 bit	R	00h

The SIO-DI32 address has a space of 1kbyte, and the actual space used is from 00h to 62h.

4.2.2. Functional Description

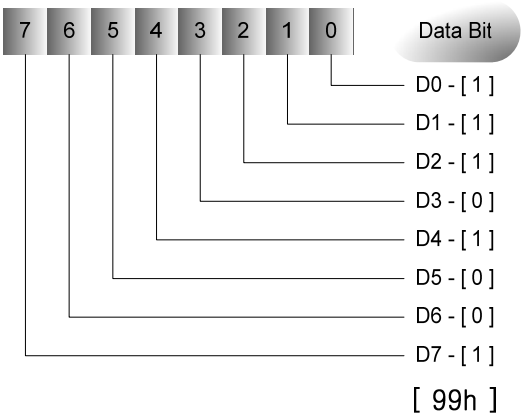
A. Preamble Register : Read Only(00h)

Preamble register arranged the combination of ‘0’ and ‘1’ and the value of ‘10110110’(0xB6) is read. This value is always read in this register address, so it is used on the automatic cognition of a module board together with the ID register.



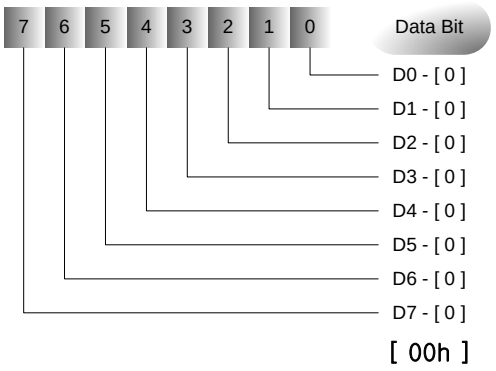
B. ID Register : Read Only(02h)

Each module has its own ID(identifier), so this information can distinguish which module board was loaded on the current baseboard. The ID of a SIO-DI32 module board is ‘10010111’(0x97h)이다.



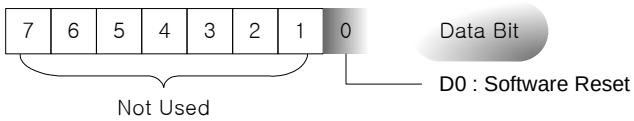
C. Version Register : Read Only(04h)

The current information on the firmware and the hardware mounted on the current module board can be captured. If you read this register address, you can also read the version of SIO-DI32.



D. Software Reset : Read Only(06h)

Using of a baseboard's reset switch or booting of a PC can initialize the state of the module board. And if you write '1' on the lowest rank bit D0 of the 06h address on the program, every register phase can be also fixed as the initial value listed in the table 5. That is, it functions as a reset switch of the baseboard.



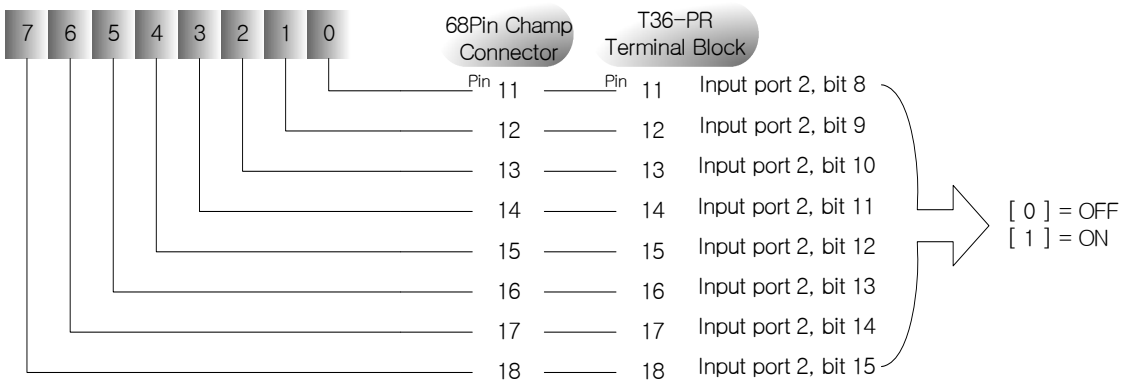
E. Input Port0 Register : Read Only(20h)

This register showed the external Input Port0's data as it is. To read the status, read the address.



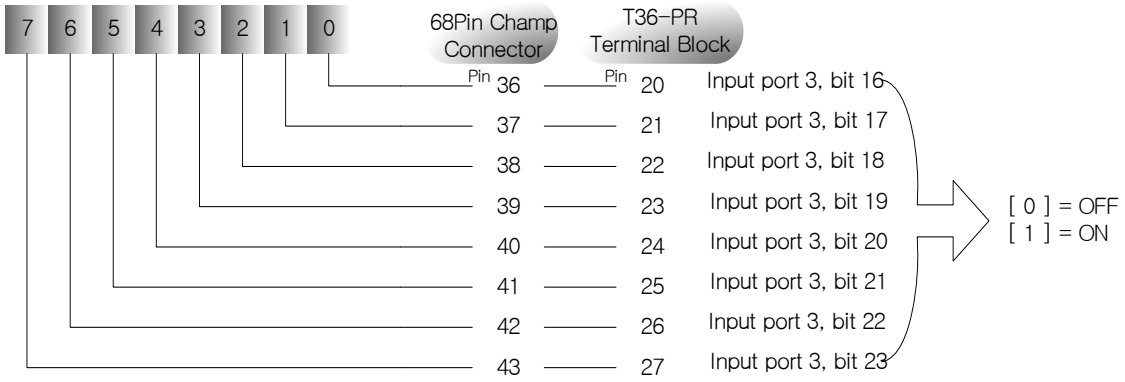
F. Input Port1 Register : Read Only(22h)

This register showed the external Input Port1's data as it is. To read the status, read the address.



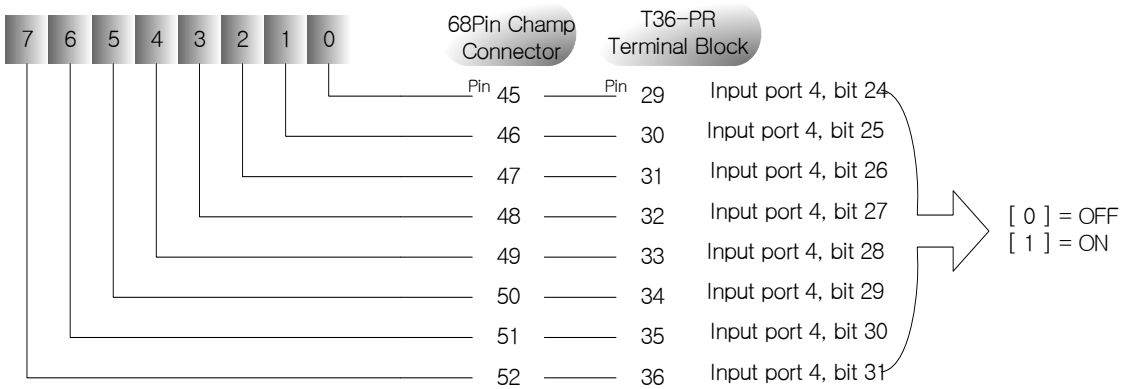
G. Input Port1 Register : Read Only(24h)

This register showed the external Input Port1's data as it is. To read the status, read the address.



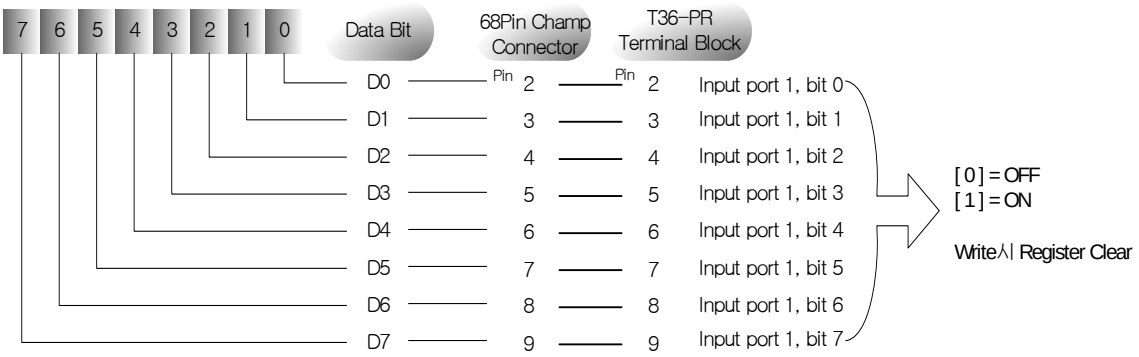
H. Input Port1 Register : Read Only(24h)

This register showed the external Input Port1's data as it is. To read the status, read the address.



I. Interrupt Flag Port1 : Read Only(30h)

Set up an interrupt on the special bit from the current module board. If the interrupt is occurred from the bit, the Interrupt Flag Port0 comes to show information about it and the interrupt occurred bit will be changed into '1'.



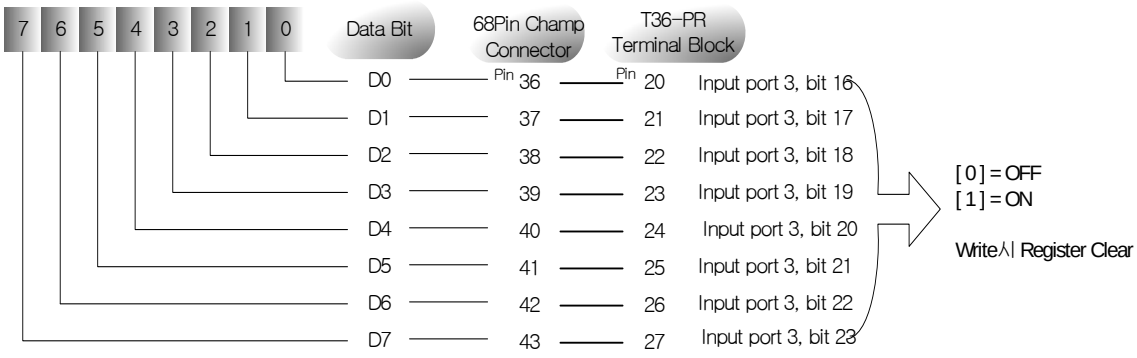
J. Interrupt Flag Port2 : Read Only(32h)

Set up an interrupt on the special bit from the current module board. If the interrupt is occurred from the bit, the Interrupt Flag Port1 comes to show information about it and the interrupt occurred bit will be changed into '1'.



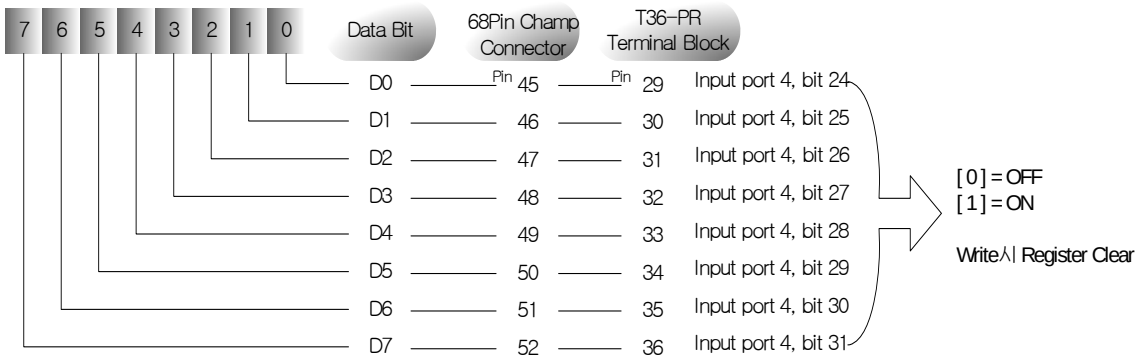
K. Interrupt Flag Port3 : Read Only(34h)

Set up an interrupt on the special bit from the current module board. If the interrupt is occurred from the bit, the Interrupt Flag Port0 comes to show information about it and the interrupt occurred bit will be changed into '1'.



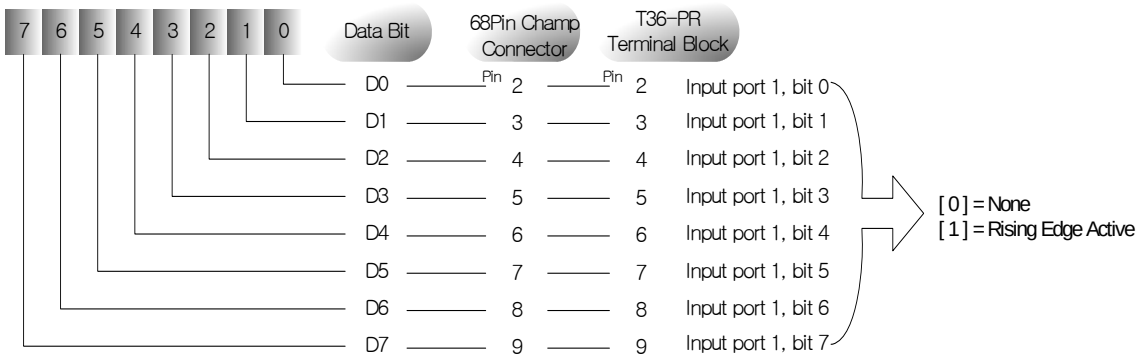
L. Interrupt Flag Port4 : Read Only(36h)

Set up an interrupt on the special bit from the current module board. If the interrupt is occurred from the bit, the Interrupt Flag Port1 comes to show information about it and the interrupt occurred bit will be changed into '1'.



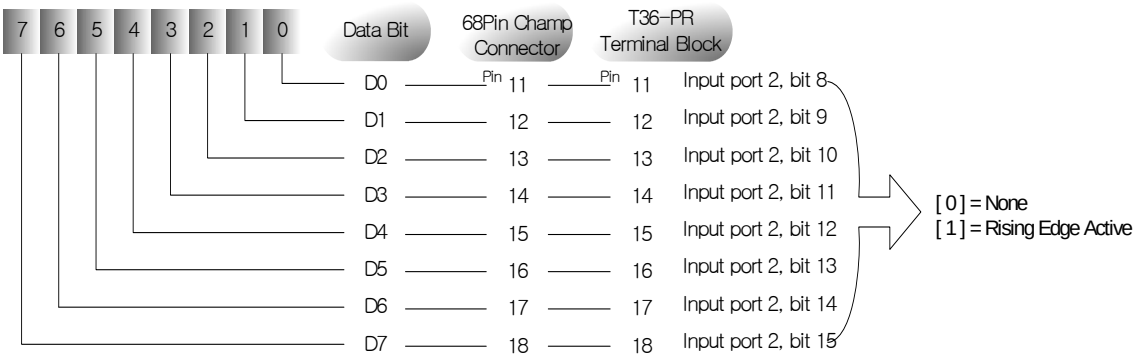
M. Interrupt Rising edge Port1 : Read and Write(40h)

This register is set to detect interrupt from the rising edge of a bit the user wants. If the Interrupt Rising edge Port1 Register bit is '1' and the input of a bit equivalent to the Input Port0 Register is changed from '0' to '1', the appropriate bit of the ↗ Interrupt Flag Port1 is set as '1', and the interrupt occurs. (Only if interrupt is enabled)



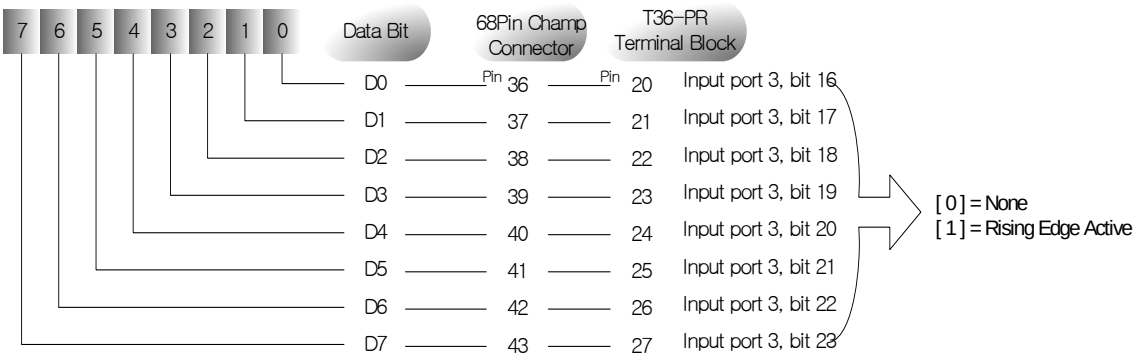
N. Interrupt Rising edge Port2 : Read and Write(42h)

This register is set to detect interrupt from the rising edge of a bit the user wants. If the Interrupt Rising edge Port2 Register bit is '1' and the input of a bit equivalent to the Input Port1 Register is changed from '0' to '1', the appropriate bit of the ⌐↑ Interrupt Flag Port2 is set as '1', and the interrupt occurs. (Only if interrupt is enabled)



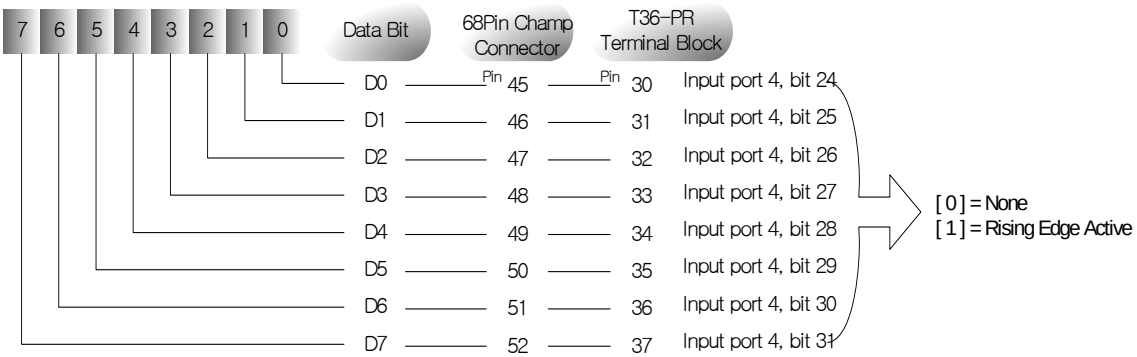
O. Interrupt Rising edge Port3 : Read and Write(44h)

This register is set to detect interrupt from the rising edge of a bit the user wants. If the Interrupt Rising edge Port3 Register bit is '1' and the input of a bit equivalent to the Input Port1 Register is changed from '0' to '1', the appropriate bit of the ⌐↑ Interrupt Flag Port3 is set as '1', and the interrupt occurs. (Only if interrupt is enabled)



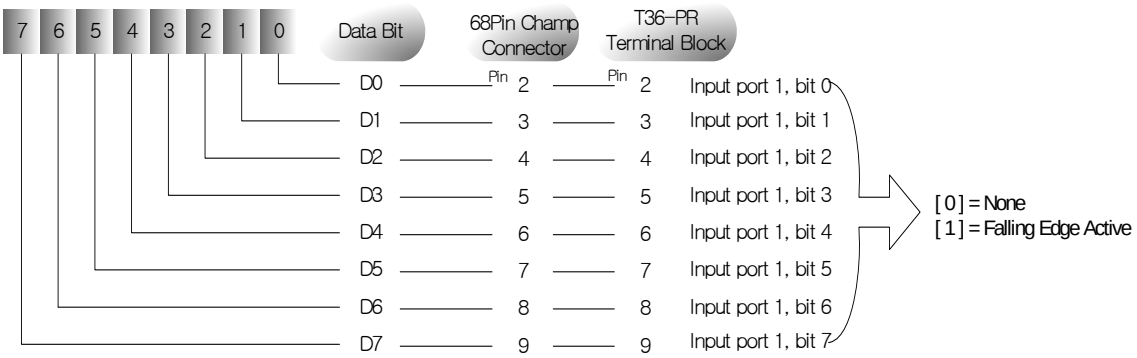
P. Interrupt Rising edge Port4 : Read and Write(46h)

This register is set to detect interrupt from the rising edge of a bit the user wants. If the Interrupt Rising edge Port4 Register bit is '1' and the input of a bit equivalent to the Input Port1 Register is changed from '0' to '1', the appropriate bit of the ↗ Interrupt Flag Port4 is set as '1', and the interrupt occurs. (Only if interrupt is enabled)



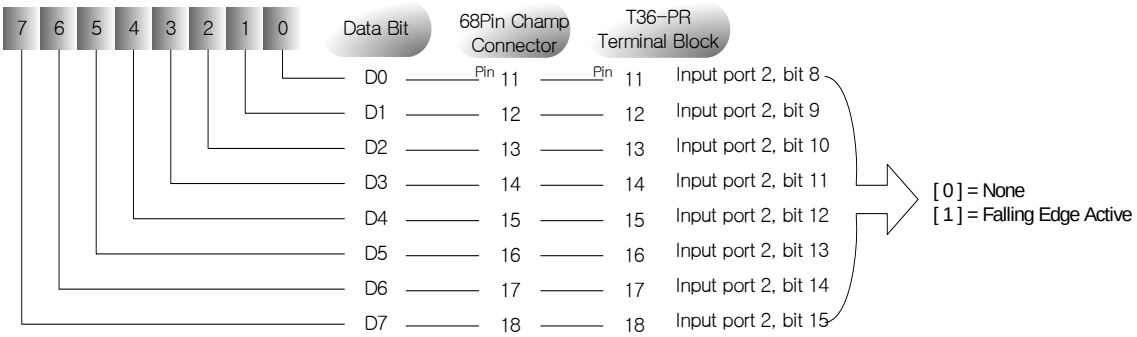
Q. Interrupt Falling edge Port1 : Read and Write(50h)

This register is set to detect interrupt from the falling edge of a bit the user wants. If the Interrupt Falling edge Port1 Register bit is '1' and the input of a bit equivalent to the Input Port0 Register is changed from '1' to '0', the appropriate bit of the ↘ Interrupt Flag Port1 is set as '1', and the interrupt occurs. (Only if interrupt is enabled)



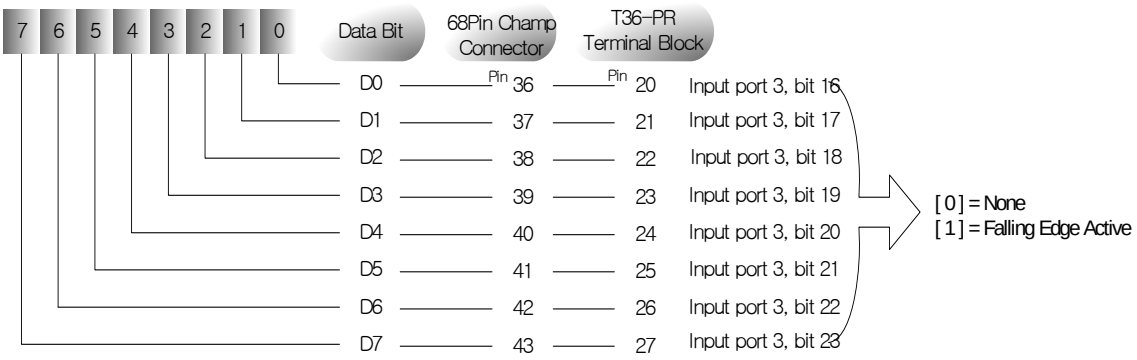
R. Interrupt Falling edge Port2 : Read and Write(52h)

This register is set to detect interrupt from the falling edge of a bit the user wants. If the Interrupt Falling edge Port2 Register bit is '1' and the input of a bit equivalent to the Input Port1 Register is changed from '1' to '0', the appropriate bit of the ↘ Interrupt Flag Port1 is set as '1', and the interrupt occurs. (Only if interrupt is enabled)



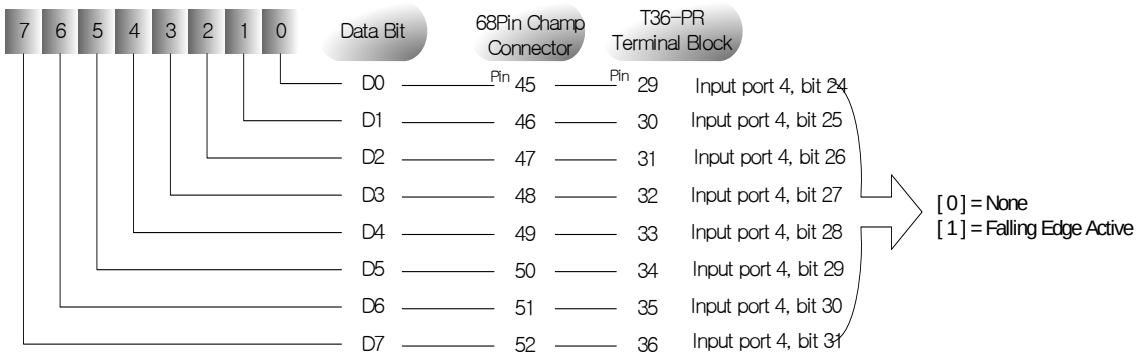
S. Interrupt Falling edge Port3 : Read and Write(54h)

This register is set to detect interrupt from the falling edge of a bit the user wants. If the Interrupt Falling edge Port3 Register bit is '1' and the input of a bit equivalent to the Input Port0 Register is changed from '1' to '0', the appropriate bit of the ↘ Interrupt Flag Port3 is set as '1', and the interrupt occurs. (Only if interrupt is enabled)



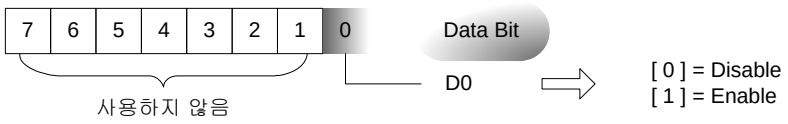
T. Interrupt Falling edge Port4 : Read and Write(56h)

This register is set to detect interrupt from the falling edge of a bit the user wants. If the Interrupt Falling edge Port4 Register bit is '1' and the input of a bit equivalent to the Input Port0 Register is changed from '1' to '0', the appropriate bit of the ↘ Interrupt Flag Port4 is set as '1', and the interrupt occurs. (Only if interrupt is enabled)



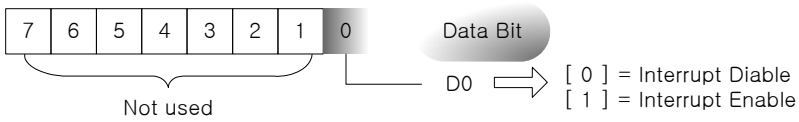
U. Interrupt Enable Control : Read and Write(60h)

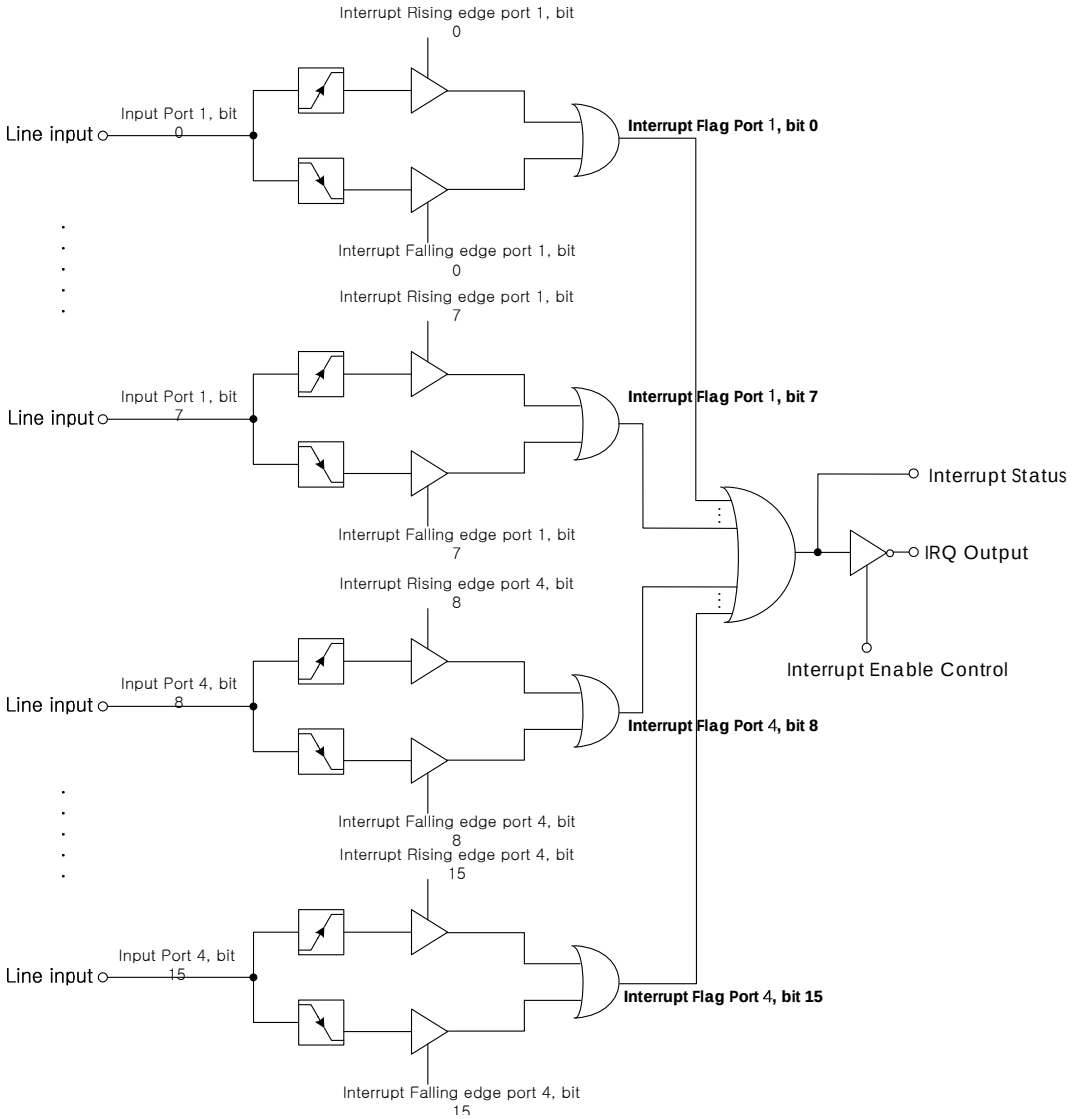
It has a function to enable/disable the whole interrupt. It is disabled when the register is '0' and enabled when it is '1'. A flag is generated according to the setting of Interrupt Rising edge Register and Interrupt Falling edge Register described above, and then it finally makes Interrupt (IRQ). Just for reference, a rising edge and a falling edge can be detected simultaneously if interrupt rising edge register and interrupt falling edge register are simultaneously set and used at the same bit.



V. Interrupt Status : Read Only(62h)

If any interrupt is occurred on the current module board, the Interrupt Status Register bit 0 will be turned into '1'. And if Interrupt Flag Port is all cleared, the Interrupt Status will be automatically cleared, too.





Picture 9. Flag Processing Block

For the full understanding, the interrupt process is illustrated as above. In picture 9, the line input is the data read from 20h and 22h, and selecting a change (rising edge, falling edge) in this data can generate interrupt. A rising edge and a falling edge can be chosen simultaneously. In this case, they are not sensed at the same time and interrupt is occurred according to the previously entered signal's status.

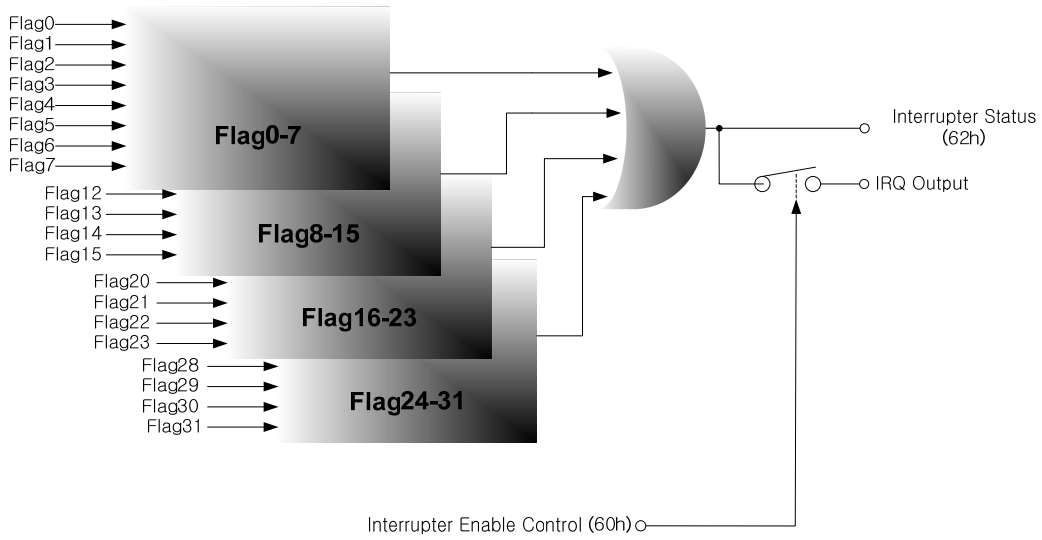
The interrupt will be occurred at the next signal, too. For instance, suppose the user selected both rising edge and falling edge. Interrupt occurs in IRQ when the falling edge signal enters to the input and it occurs again in IRQ when the rising edge signal enters to the input again.

Interrupter Enable Control (60h) should be activated to generate interrupt. That is, Interrupter Enable Control has a function to control the whole interrupt.

If interrupt is occurred, Interrupter Status (62h) should clear information on the current Flag Register to sense the next interrupt. In order to find out if any interrupt is occurred in the Flag Register, read Interrupter Status Register first. If the Flag Register is '1', clear all the current Flag Register and then wait for the next interrupt.

.

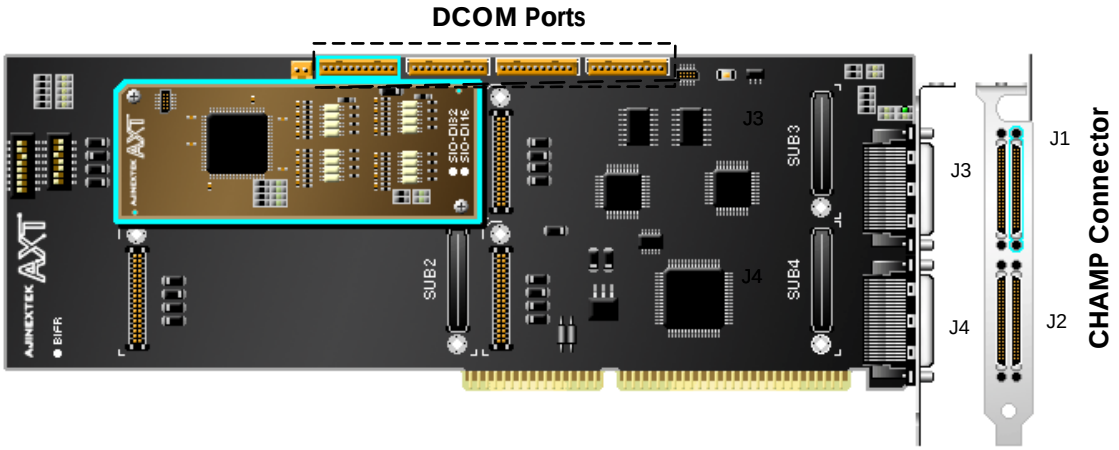
Picture 9 is simply expressed in picture 10.



Picture 10. Simplified Flag Processing and Interrupt Processing Block

4.3.Pin Array and Description of SIO-DI32 Module Board Champ Connector

SIO-DI32 is a digital input board having 32 points input, and it is electrically isolated with the external power connected through the connector. When you connect the external device and the terminal block, be sure to refer to the following table.



Picture 11. In Case of Loading SIO-DI32 on the BIFR's SUB1

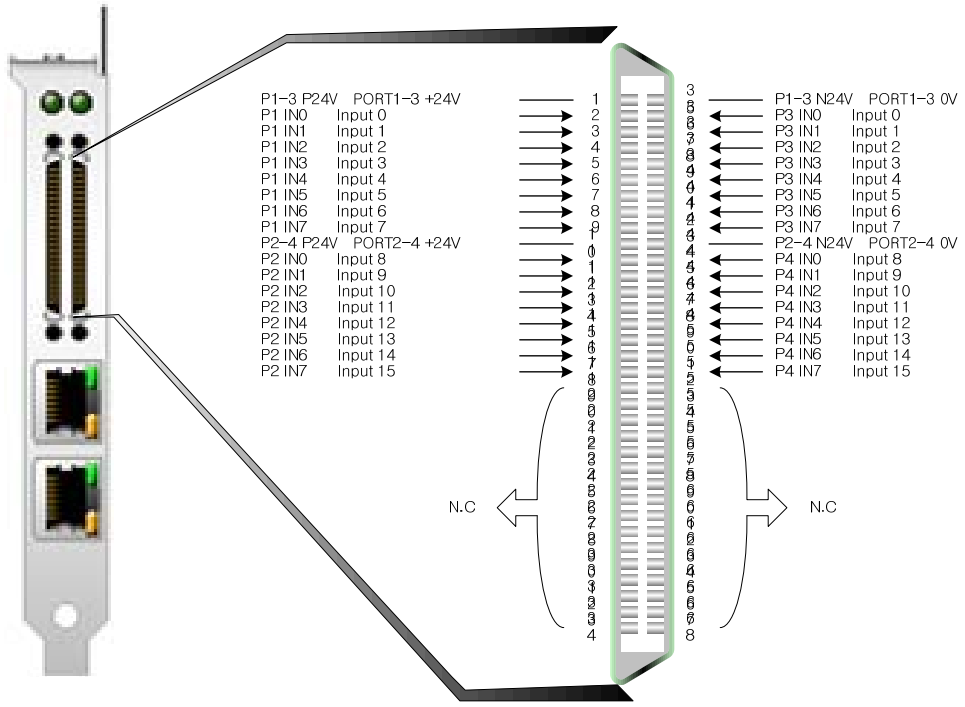
Picture 11 shows the case loaded with SIO-DI32 on the SUB1/CPU module socket of the BIFR board. The external IO interface division connected to the outside through the special or general purpose cable is composed of a 68pin CHAMP connector and a DCOM Port, a distributed master module connector, and each 68pin connector and DCOM Port are connected with the SUB module. DCOM Port is used when AnyField Series products are loaded, so this manual will not make the subject here.

Table 6. 68Pin CHAMP Connector Pin Array When SIO-DI32 Module is Loaded

J1	Pin Name	Dir.	Description	J1	Pin Name	Dir.	Description
1	P1-3_P24V		+24V Power of PORT1-3	35	P1-3_N24V		0V Power of PORT1-3
2	P1_IN0	I	Inport 1, bit0	36	P3_OUT0	I	Inport 3, bit16
3	P1_IN1	I	Inport 1, bit1	37	P3_OUT1	I	Inport 3, bit17
4	P1_IN2	I	Inport 1, bit2	38	P3_OUT2	I	Inport 3, bit18
5	P1_IN3	I	Inport 1, bit3	39	P3 OUT3	I	Inport 3, bit19
6	P1_IN4	I	Inport 1, bit4	40	P3_OUT4	I	Inport 3, bit20
7	P1_IN5	I	Inport 1, bit5	41	P3_OUT5	I	Inport 3, bit21
8	P1_IN6	I	Inport 1, bit6	42	P3_OUT6	I	Inport 3 bit22
9	P1_IN7	I	Inport 1, bit7	43	P3_OUT7	I	Inport 3, bit23
10	P2-4_P24V	I	+24V Power of PORT2-4	44	P2-4_N24V	I	0V Power of PORT2-4
11	P2_IN0	I	Inport 2, bit8	45	P4_OUT0	I	Inport 4, bit24
12	P2_IN1	I	Inport 2, bit9	46	P4_OUT0	I	Inport 4, bit25
13	P2_IN2	I	Inport 2, bit10	47	P4_OUT0	I	Inport 4, bit26
14	P2_IN3	I	Inport 2, bit11	48	P4_OUT0	I	Inport 4, bit27
15	P2_IN4	I	Inport 2, bit12	49	P4_OUT0	I	Inport 4, bit28
16	P2_IN5	I	Inport 2, bit13	50	P4_OUT0	I	Inport 4, bit29
17	P2_IN6	I	Inport 2, bit14	51	P4_OUT0	I	Inport 4, bit30
18	P2_IN7	I	Inport 2, bit15	52	P4_OUT0	I	Inport 4, bit31
19	N.C			53	N.C		
20	N.C			54	N.C		

J1	Pin Name	Dir.	Description	J1	Pin Name	Dir.	Description
21	N.C			55	N.C		
22	N.C			56	N.C		
23	N.C			57	N.C		
24	N.C			58	N.C		
25	N.C			59	N.C		
26	N.C			60	N.C		
27	N.C			61	N.C		
28	N.C			62	N.C		
29	N.C			63	N.C		
30	N.C			64	N.C		
31	N.C			65	N.C		
32	N.C			66	N.C		
33	N.C			67	N.C		
34	N.C			68	N.C		

J1, J2, J3, J4 are connected with the SUB1, SUB2, SUB3, SUB4, respectively, and they are connected to the terminal block or the external device through a cable. Therefore, 68pin CHAMP connector J1's signals are arranged as table 6 when SIO-DI32 module is loaded on the SUB1 socket. Among the 68 pins, a total of 36 pins including 32 points input and 4 power sources are used only.



Picture 7. CHAMP Connector Pin

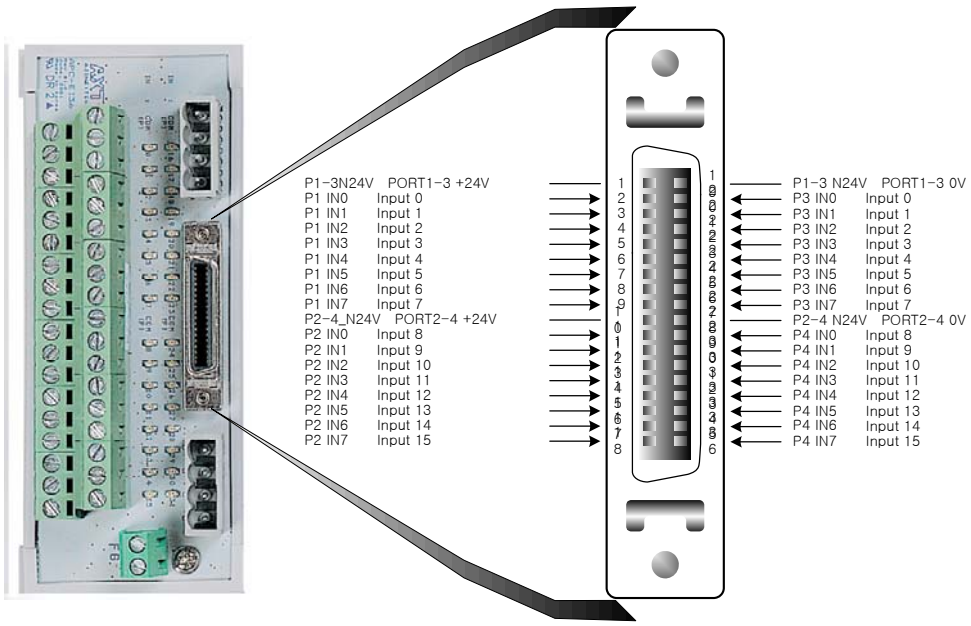
Picture 12 shows the 68 CHAMP connector's pin connected to the SIO-DI32 module.

4.4.Pin Connection and Description of SIO-DI32 Module Board Terminal Block

Table 7. Pin Connection and Description of T36-PR, APC-EI36 Terminal Block

T36-PR APC-EI36	Pin Name	Dir.	Description	T36-PR APC-EI36	Pin Name	Dir.	Description
1	P1-3_P24V		+24V Power of PORT1-3	19	P1-3_N24V		0V Power of PORT1-3
2	P1_IN0	I	Inport 1, bit0	20	P3_OUT0	I	Inport 3, bit16
3	P1_IN1	I	Inport 1, bit1	21	P3_OUT1	I	Inport 3, bit17
4	P1_IN2	I	Inport 1, bit2	22	P3_OUT2	I	Inport 3, bit18
5	P1_IN3	I	Inport 1, bit3	23	P3_OUT3	I	Inport 3, bit19
6	P1_IN4	I	Inport 1, bit4	24	P3_OUT4	I	Inport 3, bit20
7	P1_IN5	I	Inport 1, bit5	25	P3_OUT5	I	Inport 3, bit21
8	P1_IN6	I	Inport 1, bit6	26	P3_OUT6	I	Inport 3, bit22
9	P1_IN7	I	Inport 1, bit7	27	P3_OUT7	I	Inport 3, bit23
10	P2-4_P24V		+24V Power of PORT2-4	28	P2-4_N24V		0V Power of PORT2-4
11	P2_IN0	I	Inport2, bit8	29	P4_OUT0	I	Inport 4, bit224
12	P2_IN1	I	Inport 2, bit9	30	P4_OUT1	I	Inport 4, bit225
13	P2_IN2	I	Inport 2, bit10	31	P4_OUT2	I	Inport 4, bit26
14	P2_IN3	I	Inport 2, bit11	32	P4_OUT3	I	Inport 4, bit227
15	P2_IN4	I	Inport 2, bit12	33	P4_OUT4	I	Inport 4, bit28
16	P2_IN5	I	Inport 2, bit13	34	P4_OUT5	I	Inport 4, bit29
17	P2_IN6	I	Inport 2, bit14	35	P4_OUT6	I	Inport 4, bit30
18	P2_IN7	I	Inport 2, bit15	36	P4_OUT7	I	Inport 4, bit31

This table shows the pin connection and description from a viewpoint of terminal block when the SIO-DI32 module is loaded.



Picture 13. Definition of Terminal Block 36 Pin Connector Pin

5. Accessories (Optional Items)

5.1.Module Accessories and Configuration

The optional cable and terminal block used to connect the AnyDIO module and the external device are as follows.

Table 8. Terminal Block and Cable When AnyDIO Module is Loaded

Module Name	Terminal Block	Cable
SIO-DI32	APC-EI36 or T36-PR	C68/36TS
SIO-DO32P	APC-EO36 or T36-PR	C68/36TS
SIO-DB32P	APC-EB36 or T36-PR	C68/36TS
SIO-DI16 SIO-DO16T SIO-DB16T	Not in use	No private cable

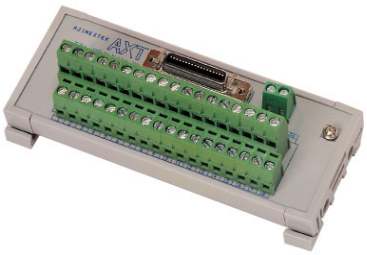
A. Cable



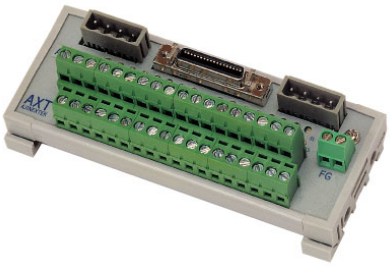
Picture 14. C68/36TS

C68/36TS, an asymmetric cable, has a Pitch (pin interval) of 0.8mm and 1.27mm. This cable connects the external connector (J1,J2,J3,J4) of the AnyBus baseboard and the terminal block one-to-one. The twist pair consists of 68 pins, such as 1 and 35, 2 and 36, …., 34 and 68. Max. 150mA of current can be flowed per each pin. Each cable is 1m, 2m, 3m long in length, and it can be supplied up to 5m long on the user's demand.

B. Terminal Block



T36-PR One-to-One Terminal Block

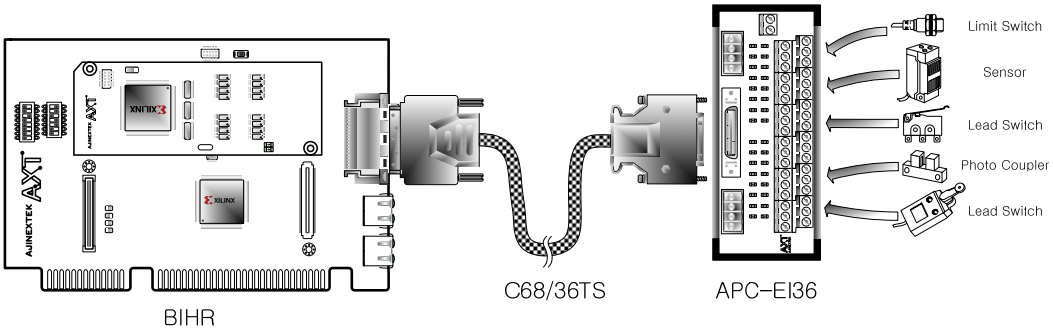


APC-EI36 Digital I/O Terminal Block

Picture15. Types of Terminal Block – One-to-One Terminal Block (left) and High Quality I/O Terminal Block (right)

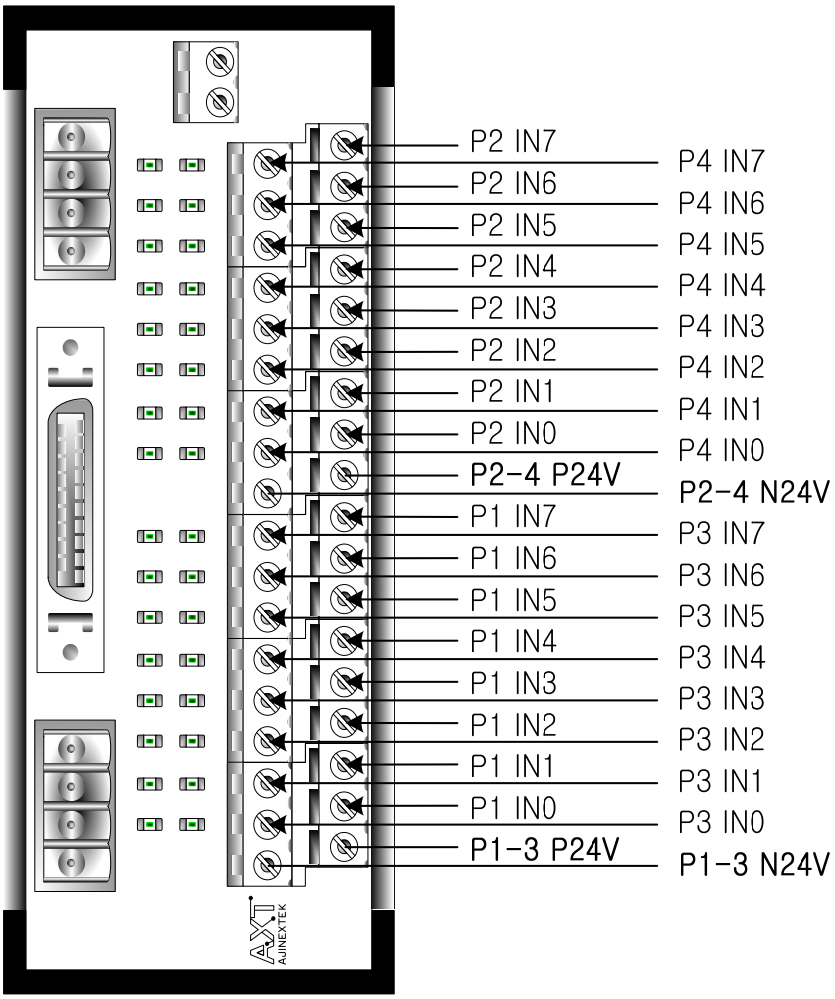
T36-PR and APC-EI36 terminal blocks are designed to be able to load on the 35mm wide slide type DIN-Rail, so it is easy to take on/off in case of installation and wiring work.

5.2.Device Connection



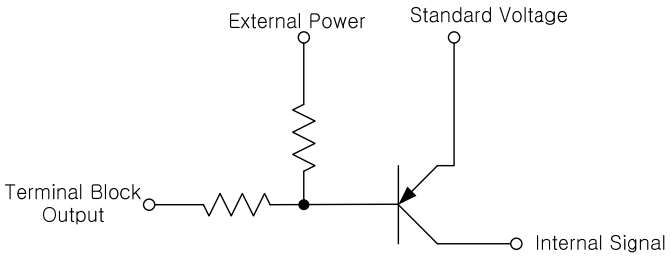
Picture 16. Connection between SIO-DI32 Module Board and the Peripheral (Terminal Board)

Picture 16 shows the connection between C68/36TS cable and APC-EI36 terminal block that are used to connect the external devices when the SIO-DI32 module board is loaded on the real BIHR board. And a diagram shows the connectable sensors to the outside of a terminal block. In addition to the sensors illustrated above, some other sensors the user wants can be used.



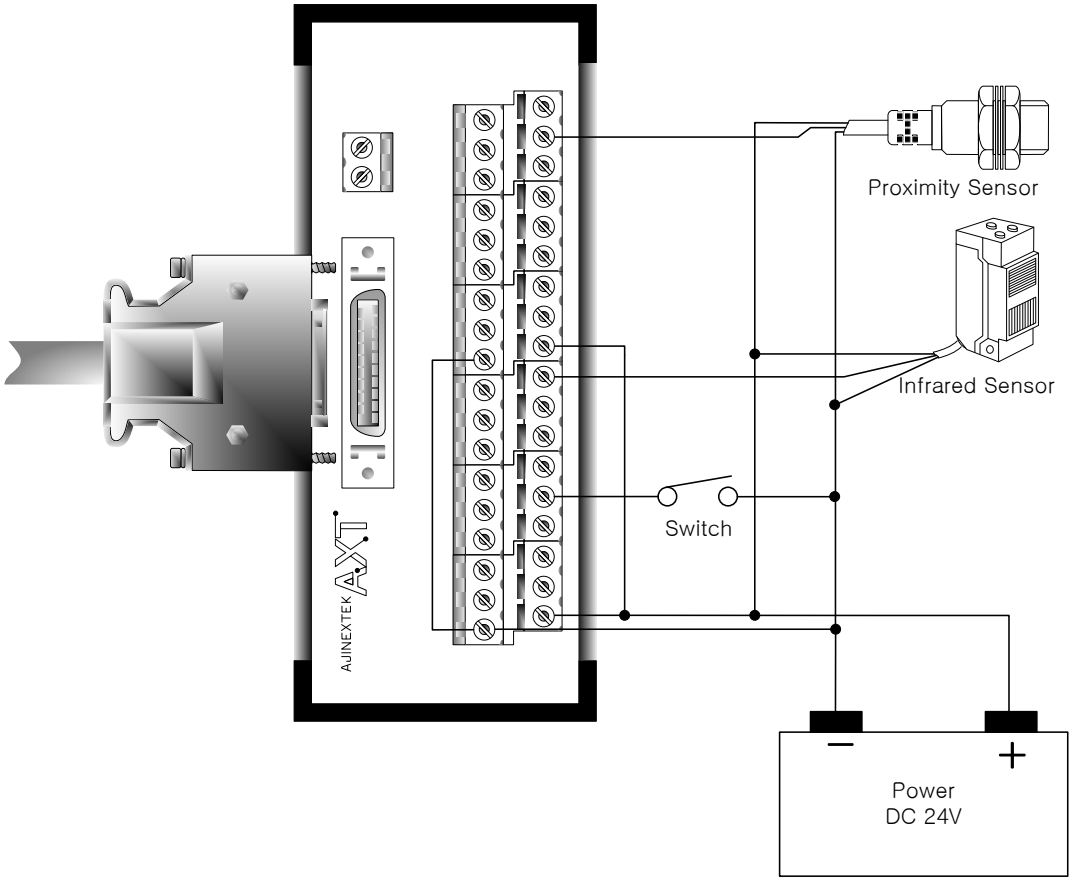
Picture 17. Description of Terminal Block Pin

Picture 17 shows the pin name and location of the terminal block.



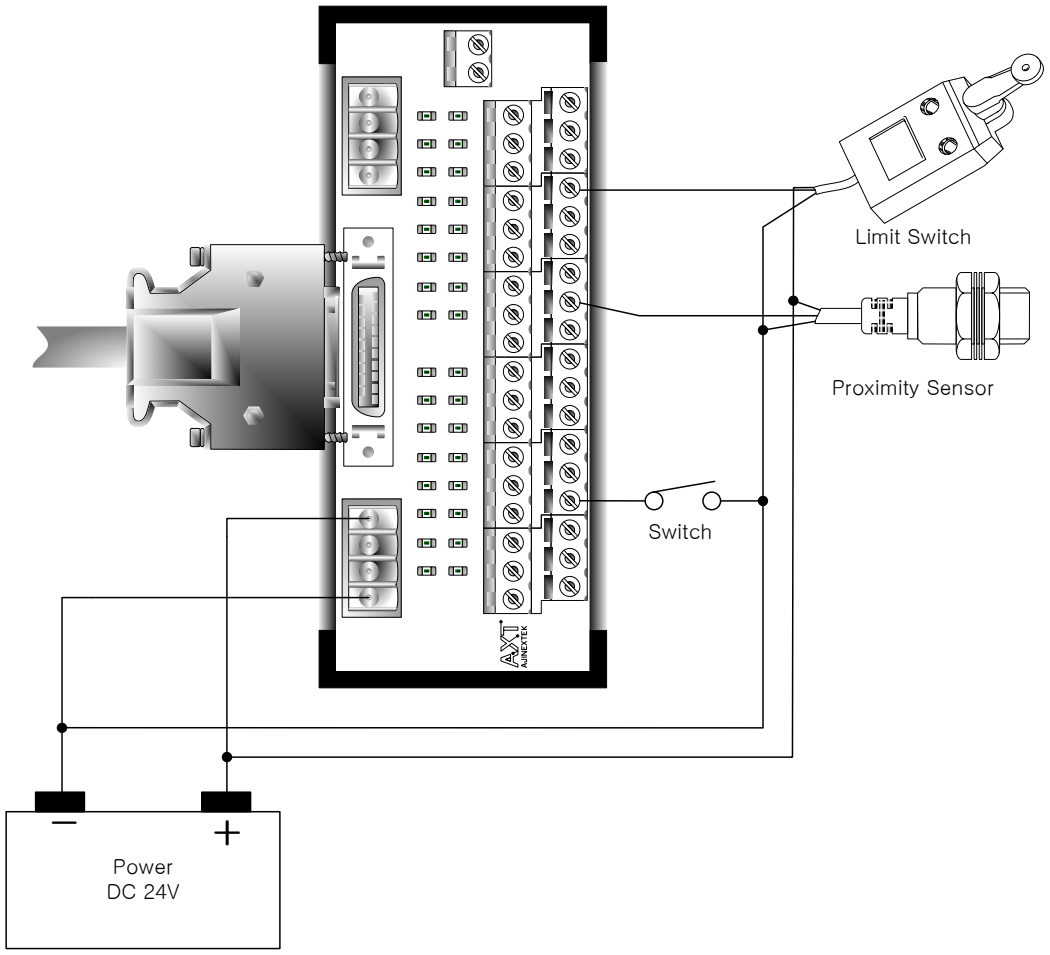
Picture 18. Input Circuit Diagram inside of Terminal Block

Picture 18 shows input circuit diagram of the terminal block.



Picture 19. Basic Configuration Using T36-PR Terminal Block

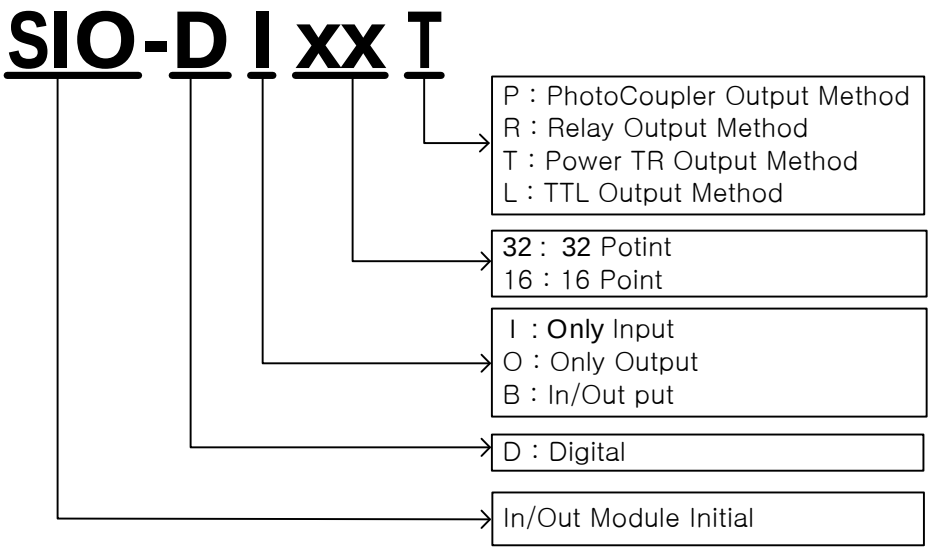
Picture 19 shows a wiring diagram between T36-PR terminal block and components. This wiring diagram shows that the switch is connected to the PORT1-IN3, the infrared sensor is connected to the PORT1-IN7 and the near-by sensor is connected to the PORT2-IN6. Input is operated by sink (negative logic).



Picture 20. Basic Configuration Using APC-EI36 Terminal Block

Picture 20 shows a wiring diagram between APC-EI36 terminal block and components. The switch is connected to the PORT1-OUT2, the near-by sensor is connected to the PORT2-IN0 and the limit switch is connected to the PORT2-IN4. APC-EI36 terminal block has two power connectors outside, so power connection between terminal blocks can be used helpfully when terminal blocks are used a lot. Input is operated by sink (negative logic).

6. Ordering Information (Family)



Picture 21. The Way of Reading AnyDIO Series Model Name

Table 9. AnyDIO Module's Product Group

Product Group	Description
SIO-DI32	(Centralized) 32 Points Digital Input Module
SIO-DI16	(Distributed) 16 Points Digital Input Module
SIO-DO32P	(Centralized) 32 Points Digital Output Module
SIO-DO16T	(Distributed) 16 Points Digital Output Module
SIO-DB32P	(Centralized) 16 Points Digital Output/ 16 Points Digital Input, Power TR (2803) Output
SIO-DB16T	(Distributed) 8 Points Digital Output/ 8 Points Digital Input, Power TR (2803) Output

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